Artificial Intelligence (AI) Project Proposal

Student Information:

- Name: Kareem Moustafa Amin (Solo Project)
- Section: 6

Department / Year:
Computer Science (CS) - 3rd Year

1. Project Title:
Smart Rescue Drone Simulator (Utility-Based Agent)

2. Project Description:

This project is a 2D interactive simulator developed using Python and Pygame. The primary objective is to design and implement an autonomous Artificial Intelligence (AI) agent acting as a "Smart Rescue Drone." The drone navigates independently within a discrete grid environment to detect and extinguish fires, rescue targets, and return to a central charging base before its battery depletes. The system operates entirely without user intervention, relying on AI search algo

3. Agent Type (Utility-Based Agent):

The system is designed as a Utility-based Agent. Unlike simple goal-based agents, this drone evaluates the "utility" (efficiency/ happiness) of its paths and actions. This specific agent type was chosen because it is highly effective in handling conflicting goals. The agent must continuously balance the trade-offs between maximizing the speed of rescue operations, preventing fire spread, and conserving battery power to ensure a safe return.

4. Agent Evaluation (PEAS Model):
To evaluate the agent's success, the PEAS components are defined as follows:

- P - Performance Measure:
Maximizing the number of lives saved, maximizing fires extinguished, and maintaining optimal battery levels to avoid crashing.

- E - Environment:
A 2D grid representing a virtual forest/city containing obstacles, dynamic fires, rescue targets, and a charging base.

- A - Actuators: Movement controls (Up, Down, Left, Right), water-drop mechanism for fires, and target-pickup mechanism.

- S - Sensors: Heat sensor (to detect fire locations), vision/camera sensor (to locate targets), and an internal battery level indicator

5. Environment Characteristics (ODESA Model):

Based on the ODESA framework, the task environment is categorized as follows:

- O - Fully Observable:

The agent has access to the complete state of the grid environment at any given time to plan its routes.

- D - Stochastic:

The next state is not entirely determined by the agent's current state and actions; fires may spread or spawn with a degree of randomness.

- E - Sequential:

Each state depends on the previous one. For example, the decision to move and consume battery life significantly affects future capabilities.

- S - Dynamic:

The environment changes while the agent is deliberating (e.g., fires continue to burn and spread over time).

- A - Single Agent:

The drone operates alone within the given environment.

- Domain:

Discrete (movement is grid-based).

6. Tools and Technologies:

- Programming Language: Python 3.

- GUI & Rendering: Pygame Library.

- AI Logic: Heuristic search algorithms (e.g., A* Search) and utility-calculation functions.